

Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands

Code:

```
geya@LenovoL480: ~/OS/Skil × + ▾
GNU nano 8.7.1
#include <stdio.h>
#include <string.h>
#include <stdlib.h>

int main() {
    char input[100];
    char output[200];
    int i = 0, j = 0;
    int double_quote = 0;

    printf("Enter text: ");
    fgets(input, sizeof(input), stdin);

    while (input[i] != '\0') {
        if (input[i] == '"') {
            double_quote = !double_quote;
        }
        else if (input[i] == '$' && input[i + 1] == 'U') {
            char *user = getenv("USER");

            if (user != NULL) {
                strcpy(&output[j], user);
                j += strlen(user);
            }

            i += 4;    // skip USER
        }
    }
}
```

```

        i += 4;    // skip USER
    }
    else {
        output[j++] = input[i];
    }

    i++;
}

output[j] = '\0';

printf("Parsed text: %s", output);

if (double_quote)
    printf("\nError: Unclosed double quote\n");
else
    printf("\nParsing successful\n");

return 0;
}
|

```

Output:

```

geya@LenovoL480:~/OS/Skill/Week 3$ ./task2
Enter text: "hello"
Parsed text: hello

Parsing successful
geya@LenovoL480:~/OS/Skill/Week 3$ ./task2
Enter text: '"Hi"
Parsed text: 'Hi

Parsing successful

```